PRANA: A Layered Framework for Sustainable AI Compute

Reframing the Energy–Water Crisis of Artificial Intelligence as an Exergy-Misplacement Problem

Shaurya Verma
Department of Computer Science and Engineering
Thapar Institute of Engineering and Technology, Patiala, India
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Abstract

The rapid scale-up of artificial intelligence has collided with three planetary constraints simultaneously: electricity, fresh water, and capital. Global data-centre electricity demand is projected to more than double from roughly 415 TWh in 2024 to about 945 TWh by 2030, while cooling-related water withdrawal already competes with agriculture and municipalities in water-stressed regions such as Texas. This paper argues that the crisis is widely misdiagnosed. From first principles, a data centre converts nearly all input electricity into low-grade heat; water consumption is a consequence of one specific design choice — evaporative heat rejection — rather than a physical requirement of computation. The sustainability problem is therefore not primarily one of consumption but of *exergy misplacement*: heat is generated where it has no value and destroyed at the cost of water and capital. Building on a survey of hardware (photonic, neuromorphic, analog in-memory), algorithmic (model routing, distillation, quantization), thermal (district-heating integration), and orbital computing directions, we propose PRANA, a five-layer framework — Physics-aligned hardware, Right-sized intelligence, Adaptive spatiotemporal scheduling, Negative-waste thermal symbiosis, and Accountability metrics — and contribute two scheduling concepts: *Follow-the-Cold routing*, which migrates deferrable workloads toward hemispheric heating demand, and the *Compute Boiler*, a facility archetype that unifies grid balancing, thermal storage, and batch computation in a single asset. We argue that treating heat as a product rather than a waste stream converts AI's largest externality into a revenue stream and materially changes the economics said to threaten an “AI bubble.”

Keywords — sustainable AI, data centres, waste-heat recovery, energy reuse factor, carbon-aware scheduling, photonic computing, neuromorphic hardware, district heating, orbital data centres

1 Introduction

Artificial intelligence has become, in the span of four years, one of the largest new categories of industrial energy demand on Earth. The International Energy Agency projects that global data-centre electricity consumption will rise from approximately 415 TWh in 2024 to roughly 945 TWh by 2030, with AI accelerator workloads accounting for the largest share of that growth [1]. A United Nations University assessment found that data centres consumed about 448 TWh in 2025 — more than all but ten countries — producing on the order of 208 million tonnes of CO_2 and drawing roughly 4.5 trillion litres of water through associated electricity generation [2]. In the United States alone, Lawrence Berkeley National Laboratory projects data centres could reach 325–580 TWh by 2028, as much as 12% of national electricity consumption [3].

The water dimension is equally stark. A single large AI facility using evaporative cooling can consume up to five million gallons per day, comparable to a town of 50,000 people [4]. Regional projections for Texas suggest that by 2030 its data centres could consume water equivalent to lowering Lake Mead by more than sixteen feet in a single year [5], and AI servers in the U.S. are projected to consume 200–300 billion gallons annually between 2024 and 2030 [6]. Li et al. estimate that global AI water withdrawal could reach 4.2–6.6 billion cubic metres per year by 2027 — four to six times Denmark's annual consumption [7].

These physical constraints have translated into economic anxiety: hyperscaler capital expenditure on AI infrastructure exceeded $355 billion in 2025 [8], and commentary increasingly asks whether the buildout is a bubble. Proposed escape routes range from incremental efficiency to placing data centres in orbit. This paper takes a different route. Rather than asking *how to feed AI's appetite*, it asks a first-principles question: *what does a data centre actually do with the energy it consumes?* The answer — it converts nearly all of it into low-grade heat — reframes the entire problem and, we argue, dissolves part of it.

The contributions of this paper are: (i) a first-principles reframing of AI sustainability as an exergy-misplacement problem; (ii) a structured survey of four solution families — hardware, algorithms, thermal integration, and orbital compute — with an honest assessment of each; (iii) the

PRANA five-layer framework that composes them; and (iv) two novel scheduling concepts, Follow-the-Cold routing and the Compute Boiler.

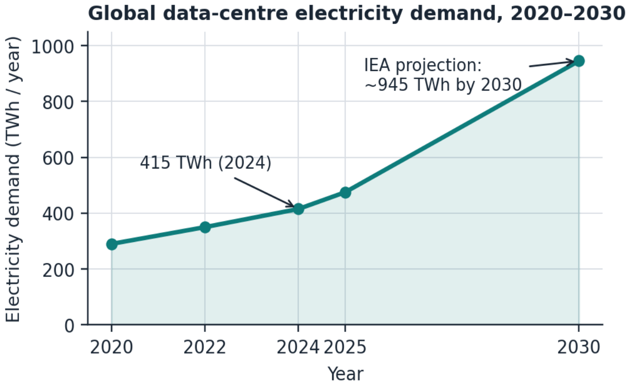

Fig. 1. Global data-centre electricity demand trajectory. Values follow the IEA Energy & AI (2025) base case [1].

2 A First-Principles Diagnosis

2.1 Computation is a heat pump with a side effect

By conservation of energy, essentially every joule of electricity entering a data centre leaves it as heat; a 1 MW IT load produces roughly 8,760 MWh of thermal energy per year at full utilization [9]. The industry's foundational metric, Power Usage Effectiveness (PUE), implicitly treats this heat as an enemy to be destroyed as cheaply as possible. Yet thermodynamically the facility is best described as an electrically driven heat source that emits computation as a side effect. The question “how do we cool data centres sustainably?” presumes the heat has no value. It does — space heating and hot water represent an enormous fraction of global final energy demand, currently served largely by burning fossil fuels.

2.2 Water use is a design choice, not physics

No step of computation requires fresh water. Water enters the picture only when operators choose evaporative heat rejection — boiling water into the atmosphere because it is the cheapest way to destroy heat. Closed-loop direct-to-chip and immersion cooling reduce direct water consumption by 70–90% [10], and Amazon reported reducing total water withdrawal in 2025 even while expanding capacity, largely by operating at higher ambient temperatures and reserving evaporative cooling for peak conditions [11]. The “AI is drinking our water” framing, while rhetorically powerful, describes a replaceable engineering decision.

2.3 The biological existence proof

The human brain performs general intelligence within a power budget of approximately 20 W — orders of magnitude below the megawatt-scale clusters used to train frontier models. Whatever the ultimate limits of silicon, biology demonstrates that the energy floor for intelligence is vastly lower than current practice. Notably, endothermic animals never “waste” metabolic heat: it is functionally integrated as thermoregulation. The biological design pattern — heat as a system function rather than a disposal problem — is precisely the pattern data-centre design has lacked.

2.4 Restating the problem

Combining these observations: the crisis is not that AI consumes too much energy, but that AI *concentrates exergy in the wrong place and then pays twice* — once in water and capital to destroy heat, and again in fossil fuel burned elsewhere to produce identical heat for buildings. We term this *exergy misplacement*. Any solution family should be judged by how directly it corrects it.

3 Survey of Solution Families

3.1 Physics-aligned hardware

A revival of analog computation — abandoned in the digital transition of the 1960s — is underway. Photonic processors perform matrix multiplication with light, and recent work integrating on-chip analog memory with photonic neuromorphic circuits reports over 26× power savings relative to conventional SRAM–DAC architectures while maintaining >90% inference accuracy [12]. Perspective analyses position photonics as a credible path to carbon-sustainable AI hardware, while cautioning that CMOS cannot simply be swapped out wholesale [13]. In parallel, neuromorphic spiking architectures and non-volatile in-memory computing attack the dominant energy cost of modern AI — data movement — and molecular-film platforms with over 16,000 conductance states hint at dramatic density gains [14]. These technologies shrink heat generation per operation; they do not, however, address where the remaining heat goes.

3.2 Right-sized intelligence

The fastest-moving lever is algorithmic. For constant model quality, inference prices have fallen by roughly 10× per year — a GPT-3-equivalent capability that cost $60 per million tokens in 2021 costs about $0.06 in 2026 [15], with per-milestone declines measured between 9× and 900× annually [16]. Gartner projects a further >90% reduction in trillion-parameter inference cost by 2030, driven by silicon efficiency, model design, and routing of routine tasks to small and domain-specific models [17]. Distillation, quantization, mixture-of-experts sparsity, and cascade routing — answering most queries with small models and escalating only when necessary — collectively mean that *most tokens never need frontier compute*. The caveat is Jevons' paradox: cheaper tokens historically induce more than proportionally more usage, so efficiency alone slows the demand curve without bending it downward [4].

3.3 Thermal symbiosis: heat as product

The most direct correction of exergy misplacement is already operating in Northern Europe. Meta's Odense facility exports heat to warm over 12,000 homes; a Fortum partnership in Finland — described as the world's largest data-centre heat-recycling project at up to 350 MW thermal — will supply roughly 40% of district heating for a metropolitan area of 250,000 people during the 2025–2026 heating season [9]. Critically, the newest AI hardware makes this *easier*: liquid-cooled high-density racks produce coolant at 55–65 °C, hot enough for direct injection into fourth-generation district heating networks without heat pumps, worth an estimated €25–35 million per year for a 100 MW campus adjacent to a network [18]. Regulation is converging on the same point: Germany's Energy Efficiency Act mandates a minimum Energy Reuse Factor (ERF) of 10% for new data centres from July 2026, rising to 20% by 2028, and the EU Energy Efficiency Directive requires waste-heat feasibility assessment for facilities above 1 MW [19]. Analyses suggest European data-centre waste heat could supply 300–350 TWh of heating per year by 2035 — on the order of 10% of EU space- and water-heating demand [20].

3.4 Orbital compute: the honest assessment

Space-based data centres attack the problem from the supply side: in a suitable sun-synchronous orbit a solar panel can be up to eight times more productive than on Earth with near-continuous illumination, cooling is achieved radiatively without water, and no community hosts the facility [21]. Google's Project Suncatcher plans prototype TPU satellites for 2027, and Starcloud launched an NVIDIA H100 to orbit in November 2025 [22]. The economics, however, are unforgiving: Google's own analysis requires launch costs to fall below roughly $200/kg — against $1,500–2,900/kg today — a threshold plausibly reached only around 2035 with very high launch cadence [21, 23]. Radiation-limited hardware lifetimes of five to six years compound the risk, and every assumption must hold simultaneously for the business case to close [24]. Orbital compute is therefore best understood as a credible 2035 supply expansion, not a 2026 sustainability strategy — and it does nothing to correct terrestrial exergy misplacement in the interim.

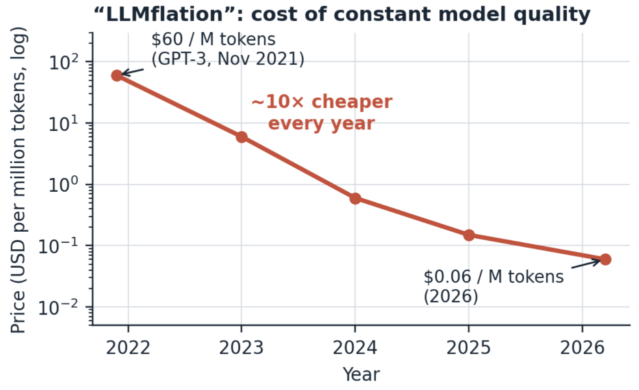

Fig. 2. Inference price for constant (GPT-3-class) quality has fallen ~1,000× in under five years [15, 16].

3.5 Cooling technology and heat reusability

Cooling method	Direct water use	Outlet temp.	DH-ready?
Evaporative (air)	Very high	25–35 °C	Heat pump req.
Direct-to-chip liquid	Low (closed loop)	50–60 °C	Near-direct
Immersion (1-phase)	Near zero	> 60 °C	Direct
Two-phase refrigerant	Near zero	up to 90 °C	Direct

Table 1. Cooling architecture determines both water consumption and the temperature — hence resale value — of waste heat [9, 10, 18].

4 The PRANA Framework

No single family above suffices: hardware efficiency is absorbed by demand growth, routing reduces cost but not placement, heat reuse requires the right geography, and orbit is a decade away. We therefore propose composing them as a stack in which each layer multiplies the layers beneath. We name the framework PRANA — in Sanskrit, the breath that animates a body — because its organizing principle is circulation rather than disposal: energy should flow *through* AI infrastructure into human use, not terminate at a cooling tower.

P — Physics-aligned hardware. Adopt the computing substrate with the lowest joules per operation available at each deployment tier: photonic and analog in-memory accelerators as they mature, liquid or immersion cooling as the default thermal interface, and inference-specialized silicon for serving workloads.

R — Right-sized intelligence. Route every query to the smallest model that meets its quality bar; reserve frontier models for genuinely hard reasoning. Distillation and quantization make the small tier continuously cheaper; cascade routing makes the choice automatic.

A — Adaptive spatiotemporal scheduling. Exploit the single most under-used property of AI workloads: much of the fleet — training, batch inference, evaluation — is deferrable in time and movable in space at the speed of light. Schedule against three live signals simultaneously: grid carbon intensity, regional water stress, and *heat demand*.

N — Negative-waste thermal symbiosis. Site and plumb facilities so that rejected heat is a product with a customer — district heating, industrial pre-heat, greenhouses, desalination — turning the ERF from a compliance number into a revenue line.

A — Accountability metrics. Replace PUE-centric reporting with per-token disclosure of energy, water, and reused heat. We propose *ERF-per-token* as the headline metric: the fraction of the energy behind each token that went on to serve a second human purpose.

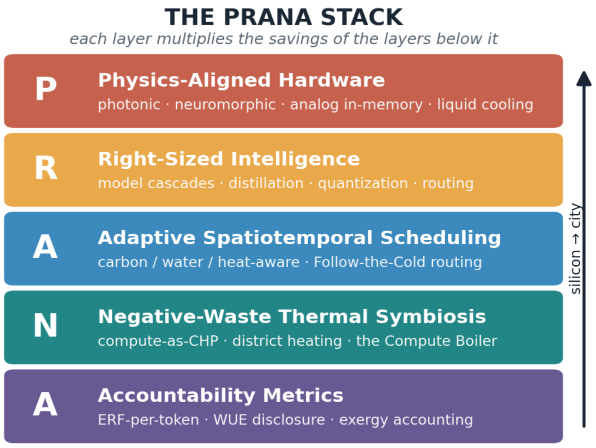

Fig. 3. The PRANA stack. Lower layers reduce heat generated per unit of intelligence; upper layers place and monetize the heat that remains.

4.1 Proposal I: Follow-the-Cold routing

Carbon-aware schedulers already implement “follow-the-sun” policies, chasing renewable supply. We propose the complementary inversion: *follow the cold*. Heating demand is among the most predictable loads on Earth, and it is hemispherically anti-correlated — January in Finland is July in Chile. A planetary scheduler holding a portfolio of heat-integrated sites in both hemispheres can migrate deferrable workloads toward wherever waste heat currently displaces fossil heating, at a round-trip latency cost of ~100–200 ms that batch workloads do not perceive. Under Follow-the-Cold, the effective carbon intensity of a training run is reduced not only by cleaner electricity but by the *avoided emissions of the boilers it replaces* — a credit existing schedulers do not model. Obstacles are non-trivial: data sovereignty and export-control regimes constrain where model weights may reside, checkpoint migration has bandwidth costs, and heating demand is seasonal while compute supply is constant. We suggest the framework be evaluated first on evaluation and distillation workloads, which are weight-static and embarrassingly parallel.

4.2 Proposal II: the Compute Boiler

The second proposal fuses two forgotten technologies with one modern one: the night-storage heater (which banked cheap off-peak electricity as heat in ceramic bricks), combined heat-and-power economics (nineteenth-century power stations sold their steam), and the interruptible GPU cluster. The seam joining them is that *batch computation and heating demand are both time-shiftable*. A Compute Boiler is a modest facility (1–50 MW) embedded in a district-heating network that (i) preferentially draws power when the grid is over-supplied with renewables, (ii) performs deferrable AI work, (iii) stores the resulting 55–65 °C heat in inexpensive water-tank thermal storage, and (iv) releases it to the network on the evening demand peak. One asset thereby sells three products — grid flexibility, heat, and computation — against one energy purchase. Because thermal storage costs orders of magnitude less per kWh than electrochemical batteries, the Compute Boiler monetizes renewable curtailment that batteries cannot economically absorb. For heating-dominated economies with young cloud markets — Northern Europe, Canada, the Himalayan states of India — this archetype suggests municipal compute infrastructure could be financed substantially by heat revenue.

5 Discussion

Does this dissolve the bubble question? Partially. The bubble thesis rests on capital expenditure racing ahead of revenue while input costs (power, water, community opposition) rise. PRANA attacks both sides: layers P and R compound the ~10×/year decline in cost per unit of intelligence [15, 17], while layers A and N convert the largest externality into revenue and social licence. A facility that heats 12,000 homes is politically difficult to oppose; one that drains a Texan aquifer is not.

Limitations. This is a position paper: Follow-the-Cold and the Compute Boiler are presented as architectures with plausibility arguments, not measured deployments. Quantitative simulation — coupling a workload trace to hemispheric heating-degree-day data and electricity prices — is the natural next step, and the author intends it as future work. Embodied carbon of chip manufacturing, rare-earth supply chains, and end-of-life e-waste are acknowledged but out of scope. Heat reuse is geography-dependent: isolated desert campuses beyond 3–4 km of any heat demand cannot apply layer N and must lean harder on P, R, and A [20].

Why optimism is rational. Every input to this analysis is moving in the right direction simultaneously: joules per operation (photonics, 26× demonstrations [12]), dollars per token (~10×/year [15]), water per joule (closed-loop cooling, 70–90% [10]), and value per joule of exhaust (regulated ERF markets [19]). The pessimistic case requires all four trends to stall; the optimistic case requires only that they continue.

6 Conclusion

The sustainability crisis of AI is real in its symptoms — terawatt-hours, trillions of litres, hundreds of billions of dollars — but misdiagnosed in its cause. Computation does not consume energy; it relocates and degrades it, and today it does so in places where the resulting heat is a liability rather than an asset. The PRANA framework proposes that sustainable AI is achieved not by asking machines to want less, but by completing the circuit: physics-aligned silicon to generate less heat, right-sized models to spend fewer joules per answer, schedulers that move work to where its heat is wanted, facilities plumbed to deliver it, and metrics that reward the whole loop. The technologies for every layer exist today, several at municipal scale. What remains is the decision to compose them — and decisions, unlike launch costs, are free.

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Correspondence: Shaurya Verma, Department of CSE, Thapar Institute of Engineering and Technology, Patiala, Punjab, India.